Lab Code

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Assignment-03

Q1. Define deadlock. Explain the four necessary conditions for deadlock occurence. Compare deadlock prevention, avoidance, and detection techniques.

Ans⇒ A deadlock is a situation in an operating system where a group of processes are permanently blocked because each process is waiting for a resource that is held by another process in the group.

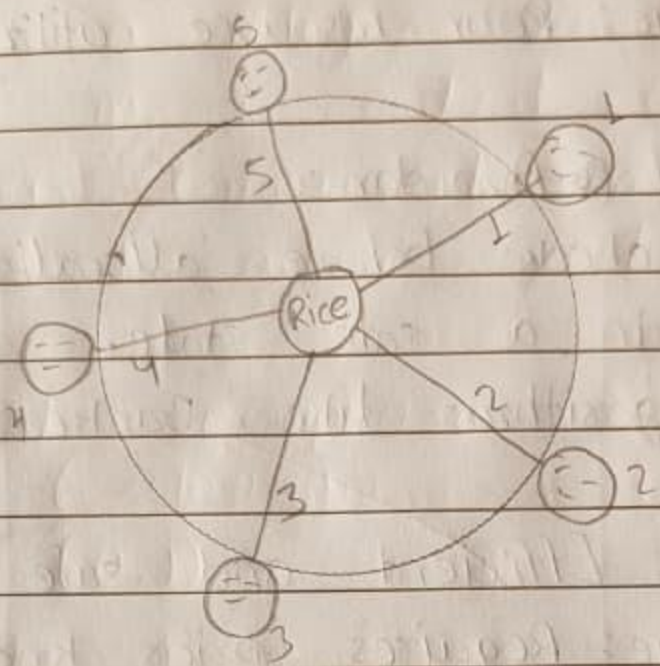
- Four Necessary conditions for Deadlock.

1. Mutual Exclusion.
   - At least one resource cannot be shared.
   - Only one process can use it at a time.
   Eg:- Printer, file, or hardware device.

2. Hold and wait.
   - A process is holding one resource and waiting for others.
   Ex:- A process holding a printer and waiting for a scanner.

3. No Preemption.
   - Resources cannot be forcibly taken away.
   - They must be ~~rest~~ released voluntarily by the process.

Ans⇒ The Dining Philosopher Problem is a classic synchronization problem in operating systems that illustrates issues like deadlock and resource sharing.



1. Thinking
2. Eating
3. No philosopher will starve
4. Philosopher may pick up only one chopstick at a
5. One philosopher can't pick up a chopstick that is already in the hand of a neighbour
6. From time to time, a philosopher gets hungry & tries to pick up the two (2) chopsticks that are closest to the philosopher.
7. When a philosopher thinks, she does not interact with colleagues.